K.R. MANGALAM UNIVERSITY

Database Management Lab Sheet

(ENCS205)

Bachelor of Technology in

Computer Science and Engineering

Submitted by:

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B. Tech CSE (Core) – 4th Semester

Section: A

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India feb-2026

Lab Sheet: 4

Lab Assignment: SQL User Management & Access Control

Title:

Implementation of **User Creation, Deletion, and Privilege Management in SQL**

Problem Statement

Design and execute SQL commands to **create users, manage their access privileges, and remove users** in a database system. This assignment focuses on **database security and access control mechanisms**.

Tasks

Part A: User Creation

1. Create a new user named `user1` with a password.
2. Create another user `user2` with limited access.

Part B: Granting Privileges

3. Grant **SELECT and INSERT** privileges on a table `Student` to `user1`.
4. Grant **ALL privileges** on a table `Course` to `user2`.
5. Provide **WITH GRANT OPTION** to `user1` and observe its effect.

Part C: Revoking Privileges

6. Revoke **INSERT permission** from `user1` on the `Student` table.
7. Revoke all privileges from `user2` on the `Course` table.

Part D: User Deletion

8. Drop/Delete user `user2` from the database.
9. Attempt to login using deleted credentials and observe the result.

Part E: Additional Commands

10. Display all users in the system.
11. Show privileges assigned to a specific user.

[QUERY]:

```sql
CREATE DATABASE lab4_db;
USE lab4_db;

CREATE TABLE Student (
id INT PRIMARY KEY AUTO_INCREMENT,
name VARCHAR(50),
age INT
);

CREATE TABLE Course (
course_id INT PRIMARY KEY AUTO_INCREMENT,
course_name VARCHAR(100)
);

CREATE USER 'user1'@'localhost' IDENTIFIED BY 'password123';
CREATE USER 'user2'@'localhost' IDENTIFIED BY 'password123';
```

Part A: User Creation

1. Create a new user named `user1` with a password.
2. Create another user `user2` with limited access

```sql
CREATE USER 'user1'@'localhost' IDENTIFIED BY 'password123';

CREATE USER 'user2'@'localhost' IDENTIFIED BY 'password123';
```

Part B: Granting Privileges

3. Grant **SELECT and INSERT** privileges on a table `Student` to `user1`.
4. Grant **ALL privileges** on a table `Course` to `user2`.
5. Provide **WITH GRANT OPTION** to `user1` and observe its effect.

```sql
GRANT SELECT, INSERT ON Student TO 'user1'@'localhost';
GRANT ALL PRIVILEGES ON Course TO 'user2'@'localhost';
GRANT SELECT, INSERT ON Student TO 'user1'@'localhost' WITH GRANT OPTION;
```

Part C: Revoking Privileges

6. Revoke **INSERT permission** from `user1` on the `Student` table.
7. Revoke all privileges from `user2` on the `Course` table.

```
REVOKE INSERT ON Student FROM 'user1'@'localhost';
REVOKE ALL PRIVILEGES ON Course FROM 'user2'@'localhost';
```

Part D: User Deletion

8. Drop/Delete user `user2` from the database.
9. Attempt to login using deleted credentials and observe the result.

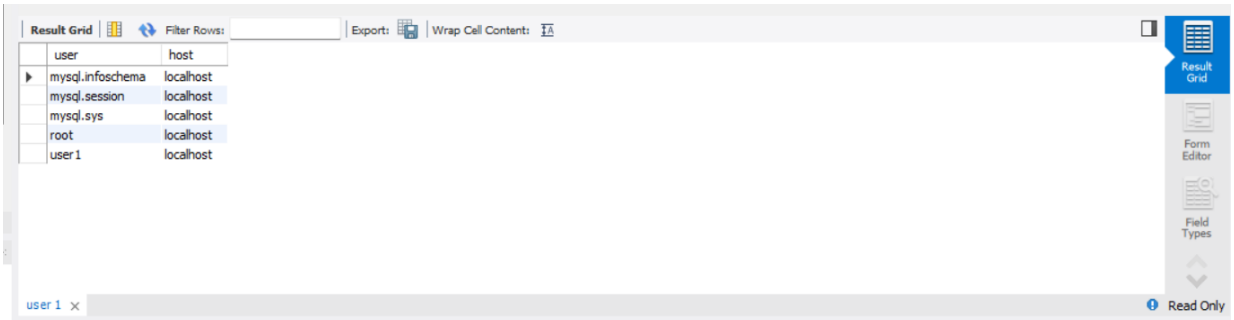
- `DROP USER 'user2'@'localhost';`

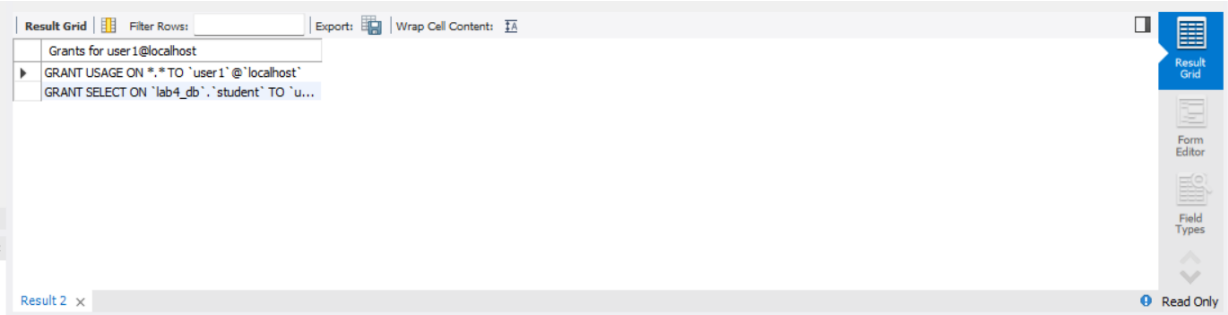
Part E: Additional Commands

10. Display all users in the system.
11. Show privileges assigned to a specific user.

- `SELECT User, Host FROM mysql.user;`
- `SHOW GRANTS FOR 'user1'@'localhost';`

[OUTPUT]:





[Observations]:

- User creation allows controlled access to the database.
- Privileges can be finely controlled using `GRANT` and `REVOKE`.
- `WITH GRANT OPTION` enables delegation of permissions.
- Revoking privileges immediately affects user capabilities.
- Deleting a user removes all access rights permanently.

[Conclusion]:

This assignment helped in understanding database security mechanisms. SQL provides powerful commands to manage user access and ensure data protection. Proper use of these commands ensures controlled and secure database operations.